```
#q1
import numpy as np

arr = np.array([1, 2, 3, 4, 5])

print("array=", arr)

print("adding 2", arr + 2)
print("multiplying by 3", arr * 3)
print("dividing by 2:", arr / 2)
```

```
array= [1 2 3 4 5]
adding 2 [3 4 5 6 7]
multiplying by 3 [ 3  6  9 12 15]
dividing by 2: [0.5 1.  1.5 2.  2.5]
```

```
#q2

import numpy as np

arr = np.array([1, 2, 3, 6, 4, 5])

print("Original array", arr)
print("Reversed array", arr[::-1])
```

```
Original array [1 2 3 6 4 5]
Reversed array [5 4 6 3 2 1]
```

```
import numpy as np

x = np.array([1, 2, 3, 4, 5, 1, 2, 1, 1, 1])

values, counts = np.unique(x, return_counts=True)
value = values[np.argmax(counts)]

print("Most frequent value", value)
print("Indices", np.where(x == value)[0])
```

```
Most frequent value 1
Indices [0 5 7 8 9]
```

```
import numpy as np

y = np.array([1, 1, 1, 2, 3, 4, 2, 4, 3, 3, 1])

values, counts = np.unique(y, return_counts=True)
value = values[np.argmax(counts)]

print("Most frequent value", value)
print("Indices", np.where(y == value)[0])
```

```
Most frequent value 1
Indices [ 0  1  2 10]
```

```
#q3
import numpy as np

arr = np.array([[10, 20, 30],
                [40, 50, 60],
                [70, 80, 90]])

print("1st row 2nd column", arr[0, 1])
print("3rd row 1st column", arr[2, 0])
```

```
1st row 2nd column 20
3rd row 1st column 70
```

```
#q4
import numpy as np

rahul = np.linspace(10, 100, 25)

print("Array", rahul)
print("Dimensions", rahul.ndim)
print("Shape", rahul.shape)
print("Total elements", rahul.size)
print("Data type", rahul.dtype)
print("Total bytes", rahul.nbytes)
```

```
arr = rahul.reshape(1, 25)

print("Reshaped array", arr)
print("Transpose", arr.T)
print("Using T", rahul.T)
```

```
Array [ 10.    13.75  17.5   21.25  25.    28.75  32.5   36.25  40.    43.75
  47.5   51.25  55.    58.75  62.5   66.25  70.    73.75  77.5   81.25
  85.    88.75  92.5   96.25 100.  ]
Dimensions 1
Shape (25,)
Total elements 25
Data type float64
Total bytes 200
Reshaped array [[ 10.    13.75  17.5   21.25  25.    28.75  32.5   36.25  40.    43.75
   47.5   51.25  55.    58.75  62.5   66.25  70.    73.75  77.5   81.25
   85.    88.75  92.5   96.25 100.  ]]
Transpose [[ 10.  ]
 [ 13.75]
 [ 17.5 ]
 [ 21.25]
 [ 25.  ]
 [ 28.75]
 [ 32.5 ]
 [ 36.25]
 [ 40.  ]
 [ 43.75]
 [ 47.5 ]
 [ 51.25]
 [ 55.  ]
 [ 58.75]
 [ 62.5 ]
 [ 66.25]
 [ 70.  ]
 [ 73.75]
 [ 77.5 ]
 [ 81.25]
 [ 85.  ]
 [ 88.75]
 [ 92.5 ]
 [ 96.25]
 [100.  ]]
Using T [ 10.    13.75  17.5   21.25  25.    28.75  32.5   36.25  40.    43.75
  47.5   51.25  55.    58.75  62.5   66.25  70.    73.75  77.5   81.25
  85.    88.75  92.5   96.25 100.  ]
```

```
#q5
import numpy as np

ucs420_rahul = np.array([[10, 20, 30, 40],
                         [50, 60, 70, 80],
                         [90, 15, 20, 35]])

print("Array")
print(ucs420_rahul)

print("Mean", np.mean(ucs420_rahul))
print("Median", np.median(ucs420_rahul))
print("Maximum", np.max(ucs420_rahul))
print("Minimum", np.min(ucs420_rahul))
print("Unique elements", np.unique(ucs420_rahul))

reshaped_ucs420_rahul = ucs420_rahul.reshape(4, 3)
print("Reshaped array")
print(reshaped_ucs420_rahul)

resized_ucs420_rahul = np.resize(ucs420_rahul, (2, 3))
print("Resized array")
print(resized_ucs420_rahul)
```

```
Array
[[10 20 30 40]
 [50 60 70 80]
 [90 15 20 35]]
Mean 43.333333333333336
Median 37.5
Maximum 90
Minimum 10
Unique elements [10 15 20 30 35 40 50 60 70 80 90]
Reshaped array
[[10 20 30]
 [40 50 60]
 [70 80 90]
 [15 20 35]]
Resized array
[[10 20 30]
 [40 50 60]]
```

```
Start coding or generate with AI.
```